CENG381 Stochastic Processes

Rate of Convergence — Practice 1

Q1 (30 points). A user is either Online (O) or Offline (I) at the start of each hour. The transition matrix is:

```
        O        I
O  [   0.6      0.4   ]
I  [   0.5      0.5   ]
```

a) Find the stationary distribution $\pi = (\pi\_O, \pi\_I)$ using the formula $\pi\_O = q/(p+q)$, $\pi\_I = p/(p+q)$, where $p = P(O \to I)$ and $q = P(I \to O)$.

b) Define $\Delta\_t = \mu\_t(O) - \pi\_O$, where $\mu\_t(O)$ is the probability of being Online at time t. Show that $\Delta\_\{t+1\} = \lambda_2 \cdot \Delta\_t$, and identify $\lambda_2 = 1 - p - q$.

c) Starting from Offline ($\mu_0(O) = 0$), how many steps until $|\Delta\_t| < 0.01$? Use $|\Delta\_t| = |\lambda_2|\hat{}t \cdot |\Delta_0|$ and solve for t.

Q2 (35 points). For the same chain as Q1, derive an explicit formula for the n-step transition probability $p\_\{OO\}^{(n)}$ using diagonalization.

a) State both eigenvalues of P. Recall that for a 2×2 stochastic matrix, $\lambda_1 = 1$ always and $\lambda_2 = 1 - p - q$.

b) Using the spectral decomposition, the n-step entry is:
$p\_\{OO\}^{(n)} = \pi\_O + \lambda_2^n \cdot \pi\_I$
Derive this formula from first principles by writing $P = \pi \cdot 1\hat{}T + \lambda_2 \cdot$ (correction matrix) and taking the nth power.

c) Compute $p\_\{OO\}^{(5)}$ exactly. What is $\lim\_\{n \to \infty\}\, p\_\{OO\}^{(n)}$? Verify it equals π_O from Q1.

Q3 (35 points). Two web servers are modeled as 2-state Markov chains on {Up (0), Down (1)}:

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Server A:   P_A = [ 0.8   0.2 ]   (p=0.2, q=0.3)
                  [ 0.3   0.7 ]

Server B:   P_B = [ 0.9   0.1 ]   (p=0.1, q=0.05)
                  [ 0.05  0.95]
```

a) Find the second eigenvalue λ_2 for each server. Which server's chain converges faster to stationarity? Why?

b) Find the stationary distribution for each server. Which server has higher long-run availability (fraction of time Up)?

c) For each server, starting from state 0 (Up), find the minimum number of steps n such that $|p_{00}^{(n)} - \pi_0| < 0.001$. Use the formula $|p_{00}^{(n)} - \pi_0| = |\lambda_2|^n \cdot \pi_1$ and compare the two servers.